Fiscal Year:	FY 2019	**Task Last Updated:**	FY 09/20/2018
PI Name:	Chahine, Georges D.Sc.		
Project Title:	Two-Phase Flow Separation and Characterization for Microgravity Applications		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	glchahine@dynaflow-inc.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	301-604-3688
Organization Name:	Dynaflow, Inc.		
PI Address 1:	10621 Iron Bridge Rd, Suite J		
PI Address 2:			
PI Web Page:			
City:	Jessup	**State:**	MD
Zip Code:	20794	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	10/21/2015	**End Date:**	10/20/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
Contact Email:	john.b.mcquillen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE change in grant number to 80NSSC18M0001 per NSSC information and J. McQuillen/GRC (Ed., 1/16/2018)		
Key Personnel Changes/Previous PI:	September 2018 report: Amir Mansouri, Ph.D., has joined the project as CoInvestigator.		
COI Name (Institution):	Wu, Xiongjun Ph.D. (Dynaflow, Inc.) Loraine, Greg Ph.D. (Dynaflow, Inc.) Mansouri, Amir Ph.D. (Dynaflow Inc.)		
Grant/Contract No.:	80NSSC18M0001 ; NNX16AB85A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Microgravity Fluid Separation: Dynaswirl® Bubble Vortex Separator," grant NNX11AO76A with new grant NNX16AB85A and title, "Two-Phase Flow Separation and Characterization for Microgravity Applications" and same Principal Investigator Georges Chahine, D.Sc. (Ed., 4/18/2016); 1/16/2018 update: New grant number 80NSSC18M0001 awarded for same title--Two-Phase Flow Separation and Characterization for Microgravity Applications (Ed., 1/16/18) The limits on the amount of liquids and gases carried to space make it imperative to recycle and reuse. On Earth, bubbles are relatively easily separable by buoyancy. In microgravity, other external forces such as a centrifugal force must be utilized to separate bubbles from liquids. Two categories of centrifugal force separators exist: one uses active rotation by mechanically spinning the tank. This is very efficient but requires shaft, bearings, and motor. In the second, the tank is fixed and the rotation is induced by eccentric injection of the mixture (Free Vortex Separator or FVS). This FVS has been proven very efficient for mid-range void fraction (50-80% gas) with an efficiency approaching 100%. However, this FVS is not as efficient for lower void fraction, and there is a need of a separator for low void fractions. We have developed a phase separator at DYNAFLOW based on a DYNASWIRL® swirl flow nozzle. It consists of two concentric cylinders, the fluid is introduced in the space between the two cylinders, and the swirling flow inside the inner cylinder is generated with the help of tangential injection slots. With this configuration one can make the vortex circulation intense enough to generate cavitation with a very stable vaporous/gaseous core even at low flow rates. Operation with the high swirl near or slightly below cavitation inception in the vortex core guarantees removal of air even from mixtures with very low void fractions. This is achieved by forcing the bubbles to expand and collect in the vortex core. These are then removed through two central exit orifices at each end of the cylinder. This configuration was tested successfully during a reduced-gravity flight and we are planning additional testing. We are further developing and experimentally testing this configuration to characterize its gas-liquid separation performance and to improve its efficiency. We are also conducting numerical simulations to help optimize the design and ultimately build a compact system for space validation on the International Space Station (ISS) using the existing Fluid Integrated Rack facility.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As a passive separator device, the DYNASWIRL® phase separator has the advantage of simplicity and dependability. It requires no moving mechanical parts and low power. Compared to other passive separator devices, which usually are not very efficient at low void fraction, the DYNASWIRL® is able to force low void fraction gas out of the liquid into the central core of the vortex by combining the effects of swirl, cavitation, and rectified gas diffusion. In addition to being an effective phase separator for space applications, it will also find wide applications in Earth applications where fast separation of low void fraction gas is of concern.
Task Progress:	During this reporting period, we have focused our efforts on working with the NASA engineering team and its contractor, Zin Technologies, to develop material and prepare for the Requirements Definition Review (RDR). This involved finalizing the requirements for each component of the system and each test point of the planned experiments in order to insure successful flight tests of the DYNASWIRL® phase separator onboard the International Space Station (ISS). Significant progress was made in the following tasks: A. Performance envelope of the system under different operating conditions: Systematic ground tests were conducted with the phase separator to obtain performance envelops with the operation parameters of the system in the planned ISS tests (maximum gas flow rate as a function of liquid flow rate within the maximum operation pressure). The generated ground performance maps corresponded to the ISS planned tests and provided guidelines for fine-tuning the planned test matrix in the ISS experiments. An important figure of merit of the separator performance is the ratio of the flow rate of gas extracted over the gas inflow rate. This was measured when all four injection slots of the DYNASWIRL® separator were used and when two of them were blocked. We also measured the gas extraction performance when the downstream pressure at the liquid exit of the phase separator was fixed at 1 atmosphere and when it was left uncontrolled. The limits of operation of the system when the pressure downstream was not controlled were obtained for different operation pressures in the upstream of separator chamber within a limit safe operation pressure. This data helped us update the test matrix with test points more suitable to evaluate the separator's performance on the ISS. B. Refined Numerical Simulations: 3D Numerical simulations were conducted by coupling our 3DYNAFS-VIS©, a two-phase flow Navier-Stokes solver to model the overall flow field of the bubbly mixture in the DYNASWIRL® phase separator, with 3DYNAFS-DSM©, a bubble tracking, and dynamics motion solver. Both components are modules of our 3DYNAFS© CFD suite and are coupled at each time step. Simulations were conducted to check the effects of the chamber size, the swirl parameter, the number and sizes of the injection slots, and different dimensions of the swirl chamber, etc. The results from the numerical simulations were used to finalize the phase separator dimensions to be fabricated for the ISS experiment. C. Updated SRD: We worked very regularly through online meetings with the engineering team members at NASA Glenn Research Center (GRC) and with Zin Technologies to finalize the SRD (Science Requirement Document) needed for the engineering work. This covered the requirements for high-speed camera visualization, lighting, bubble size measurement device, pressure and flow measurement devices, etc. We also had in person discussions at Zin Technologies site with their team. D. Updated Test Matrix: Due to the changes of the constraints imposed by the available resources for building the system and data obtained from ground tests and numerical simulations, the test matrix was updated and refined through multiple iterations to maximize the coverage of the scientific objectives within the limits of the available resources. E. Instrumentation Development for Void Fraction Measurement: Efforts were also made to provide reliable instrumentation for high void fraction measurement of the two-phase flow mixture at the inlet of the phase separator. To do so, electric impedance probes were tested extensively at different inclination angles of the test loop and the results were compared to those obtained with quick action valves to capture the gas and water and measure the void fraction. Tests were conducted at different liquid and gas flow rates and gravity directions and nice correlations were obtained between void fractions and impedance probe readings. F. Requirements Definition Review (RDR): In coordination with the NASA GRC engineering group and Zin Technologies, we prepared for and presented our approach to the Requirements Definition Review (RDR) at GRC in

	June. The review presented our preparedness and readiness along with the scientific and engineering teams, and we received feedback from the review panel for further improvements.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Abstracts for Journals and Proceedings	Wu X, Loraine G, Mansouri A, Chahine G. "Ground Studies of a DynaSwirl® Phase Separator for Space Applications." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Hsiao C-T, Mansouri A, Wu X, Chahine G. "Experimental and Numerical Study of Gravity Effects on Bubble Capture by a Line Vortex in a Phase Separator." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017